individual step will only take 30 milliseconds and in between each step, we can, for example, check the status of a digital input pin and exit the loop using the `break` statement if a condition has been met.

With all of the Stepper library functions out of the way, let's look at one more sketch in Listing 9-7 before we move on to our next and final library for this chapter. This sketch demonstrates one way to map the values from an analog sensor to the movement of the stepper motor. A potentiometer with a giant knob on it is a good choice, or maybe better yet an accelerometer that controls the movement of the stepper through the tilt of its axes, although any analog input would work as well. Using the `map()` function, we can take an analog reading and map it to the total number of steps to indicate the level of the analog input—a full reading of 1024 will turn the stepper one full revolution.

Listing 9-7. *Analog Sensor to Stepper Source Code*

```
#include <Stepper.h>

const int steps = 200;
Stepper stepper(steps, 8, 9, 10, 11);
int previous, val;

void setup() {
  stepper.setSpeed(30);
}

void loop() {
  val = map(analogRead(0), 0, 1024, 0, steps);
  stepper.step(val - previous);
  previous = val;
  delay(20);
}
```

Now that we have used libraries to display text on an LCD as well as making things move with two different kinds of motors, let's take a look at how we can use libraries to store and retrieve information.

SD Library

In the last chapter, we very briefly introduced the idea of using the program memory of the Arduino microcontroller to keep large chunks of data from taking up too much room in RAM. Instead, we could use additional external hardware to read and write to an SD memory card. SD or Secure Digital flash memory cards are the very same wafer-thin memory cards used by cameras, mp3 players, and other devices. The SD library is a relative newcomer to the standard Arduino libraries and the list of functions and associated libraries would make our conversations of the LiquidCrystal library and the `PROGMEM` functions seem like a leisurely stroll in the park in comparison. Rather than a comprehensive analysis of the library in the few remaining pages of this chapter, we are going to take a quick look at reading and writing values to the card using as few functions as we can get away with.